A number without its ground

Fingerprints for the OSPD mangrove workflow. Offline render of `fingerprints_marimo.py`, Sundarbans, resolution 9, OGC API-DGGS Annex B rHEALPix profile.

Pages follow the twelve screens of the live notebook. Interactive controls are omitted; every figure is the state the notebook opens in.

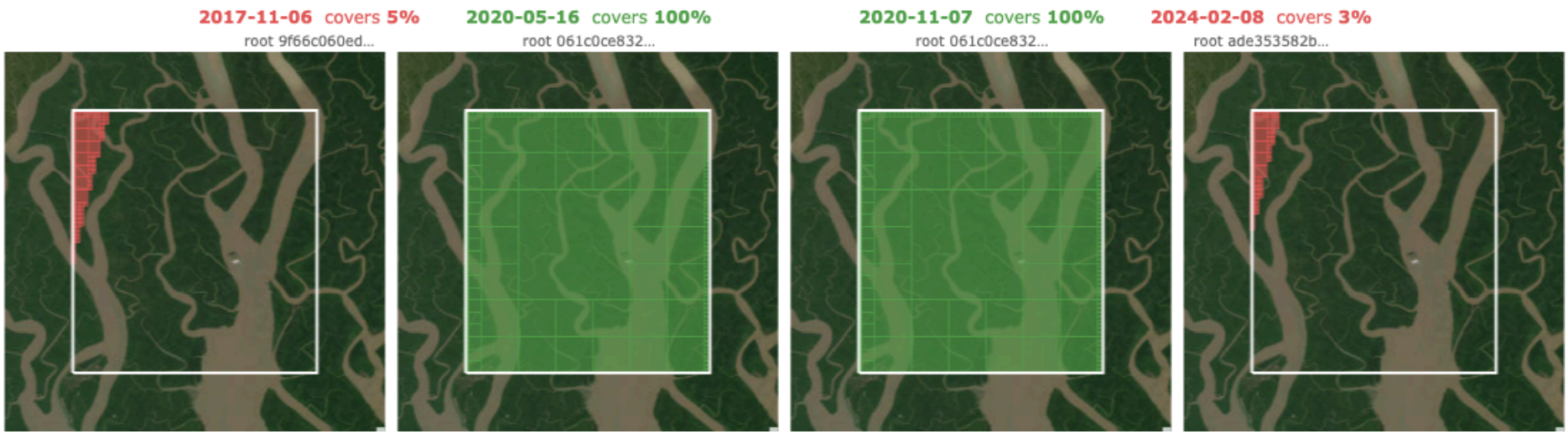
A number without its ground

> The story in one line. Give every dataset an identity that is a function of the ground it covers. Then a value carries its own caveat, scenes are chosen and compared on ground rather than on hope, change is a diff, a catalogue can answer what covers this , and a platform stops moving bytes nobody asked for.

The mangrove workflow was delivered to OGC's Open Science Persistent Demonstrator in 2025. This is what it could not say, and what one small change to its outputs makes possible.

1. What we reported in 2025, and where each number came from

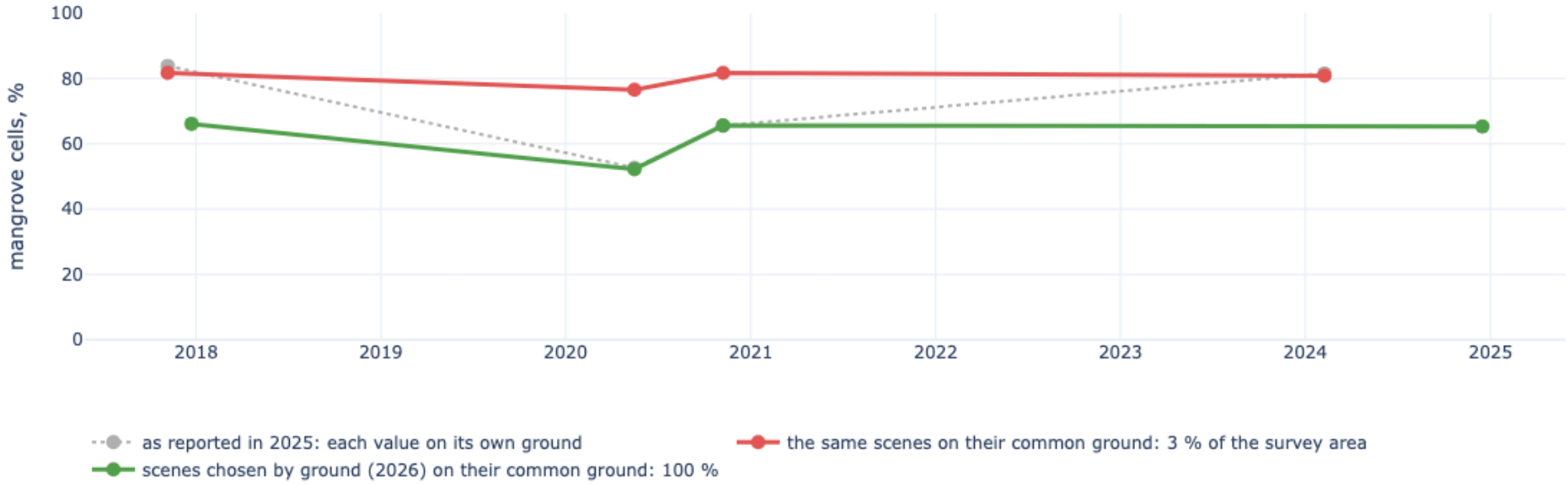
> Takeaway. The 2024-02-08 value stood on 3 % of the survey area, a sliver at a tile edge. The table could not say so. A number without its ground is not a datapoint.



date	scene	mangrove fraction %	mean biomass Mg/ha	ground it was computed on	share of the survey area
2017-11-06	S2B_45QYE_20171106_0_L2A	83.9	135.0	9f66c060ed30...	5 %
2020-05-16	S2A_45QYE_20200516_0_L2A	52.6	65.5	061c0ce8321c...	100 %
2020-11-07	S2B_45QYE_20201107_1_L2A	65.6	117.0	061c0ce8321c...	100 %
2024-02-08	S2A_45QYE_20240208_0_L2A	81.5	128.4	ade353582b21...	3 %

The last two columns are new: the root names exactly which cells the value was computed on, and the zone count is an exact area. They travel with the value wherever it is copied.

Three series, each with its ground



Grey is the 2025 table: four numbers on four pieces of ground, not a series. Its dip and recovery is the ground switching, not the forest. Red is the honest version of the same scenes, each restricted to the ground all four saw. Green is what the workflow reports now that it chooses scenes by ground. Red and green are both legitimate: different ground, different measurement, and the root is what says so.

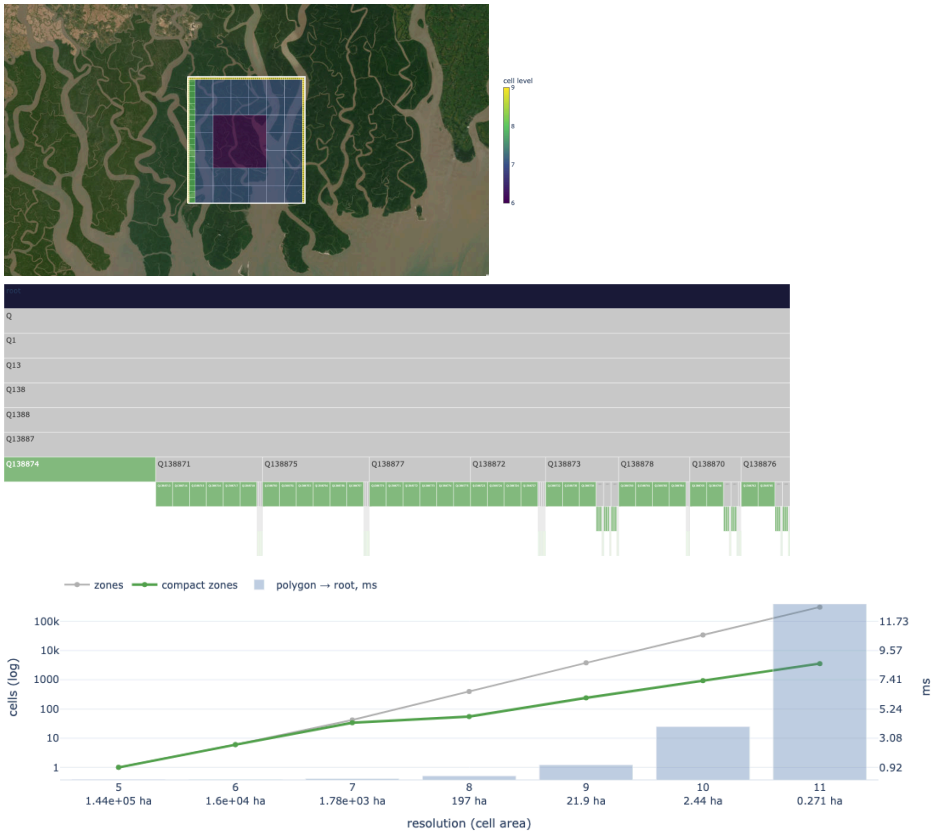
2. Naming the ground: fair cells, rolled up, hashed

> Takeaway. The survey area becomes 3,776 equal-area zones, rolled up to 240, and hashed to one root. Anyone with the same polygon, resolution and profile recomputes the same 32 bytes, in 1 ms.

zones at resolution 9
3,776
each 21.94 ha, exactly

compact zones
240
nine full children become their parent, recursively

the root
061c0ce8321c7b3d…
SHA256 over the canonical tree



240 cells drawn · drag left to see every leaf, right to see the canonical form

Grey nodes are partial and hashed from nine children; green nodes are full and hash to a constant. 99 hash evaluations produce the root, not 3,776. A region held whole and the same region spelled out leaf by leaf hash identically.

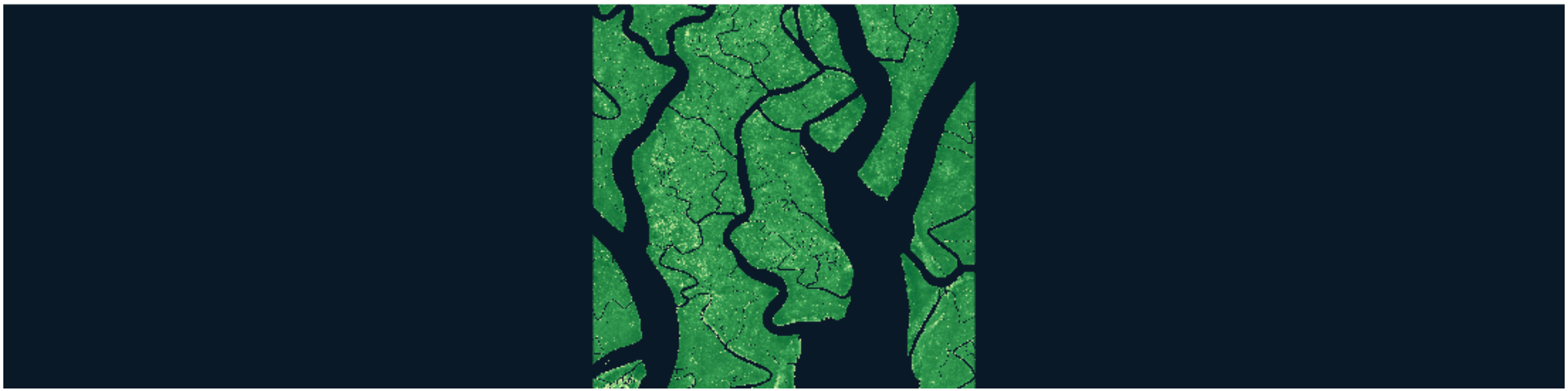
Leaf cells grow ×9 per resolution; compact zones grow with the boundary. At resolution 11 the area is 308,028 leaves but 3,588 compact zones. A continent and a field each cost one root.

3. The same ground in any encoding is one root

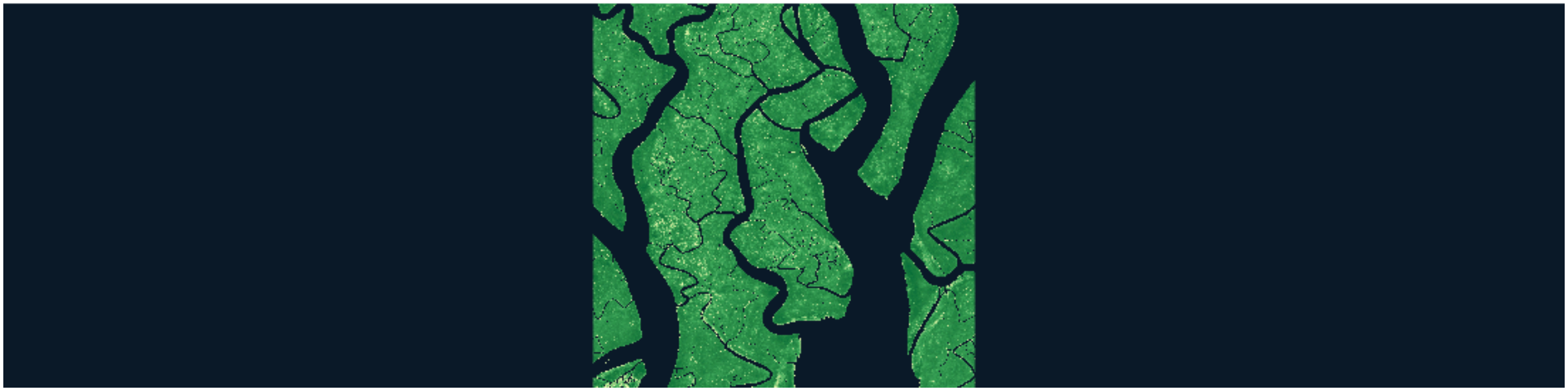
> Takeaway. The 2024-12-16 biomass map written as a tiled COG and as a chunked Zarr store: two files, one footprint, one root. Re-tile it, re-compress it, move it between object stores; the identity does not move.

root of the COG	root of the Zarr	verdict
86a1308fe0079816…	**86a1308fe0079816…**	**identical**

COG · 256² tiles · deflate · 1,413,648 bytes



Zarr · 64² chunks · 82 files



4. You only lose on entering the system

Takeaway. The same mangrove mask, two routes. Break-bulk: reprojected through twenty hops of the kind data really makes (UTM at 30 m, WGS84, Web Mercator at 60 m, a different WGS84 grid, UTM at 50 m, round again): after twenty handlings 8.3 % of the forest is in the wrong place while every area total stayed within 0.6 %, so a summary table would never notice and a change detector would report change that never happened. Container: fingerprinted once at level 11 (entry cost 0.1 % of overlap, paid once), then twenty operations: root 3c0a4223e6d9ee50… every time, 200,593 zones every time. Loss stops at one.

of the forest moved
8.3 %
after twenty reprojection hops

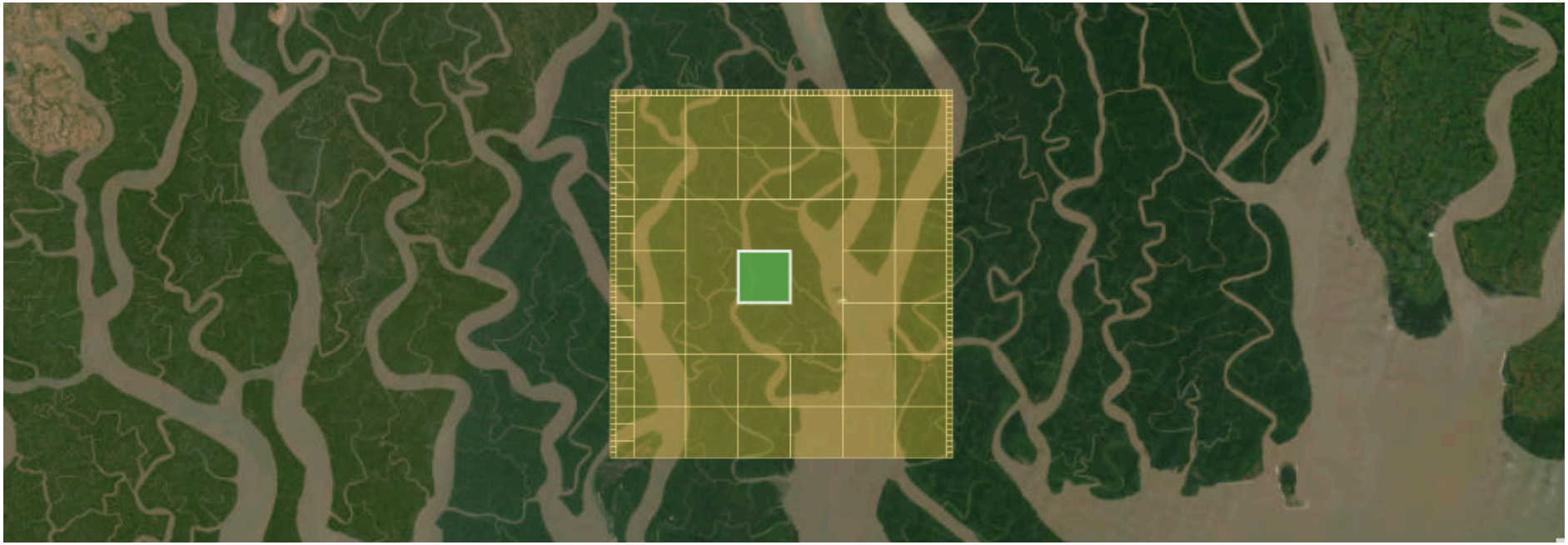
worst drift in area
0.64 %
the totals look fine

operations that changed the root
0
out of twenty



5. The root answers questions without the data

> Takeaway. Hand someone the root and they can ask is this cell inside? and get an answer they can check, for any cell, without the polygon or the raster. Absence is proven the same way as presence.



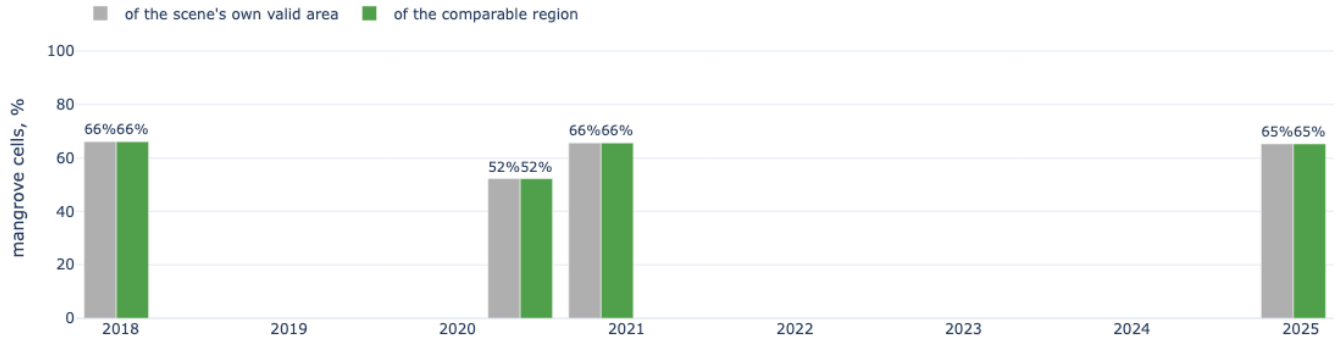
Q1388744 is inside the study area's coverage. The proof is 4,424 bytes, checks against the root (True) and fails against a forged root (False). Nobody needed the polygon, the raster or the tree to check it: only the root.

6. Four passes, one comparable region

> Takeaway. Nobody sees the whole area in one pass. Each scene's footprint is a root; the region they all saw is the intersection, computed, not assumed. And scenes are now chosen by how much of the area their footprint covers, before any download: the 2025 report's own first recommendation.







Grey bars are what each scene says about itself; green bars are the number that means the same thing on every date. 4 scenes, 1 distinct footprint roots: scenes from one Sentinel-2 tile share a root to the byte.

7. Change is a set difference, 2017-12-23 → 2024-12-16

> Takeaway. Two roots, one difference: 790 ha lost, 88 ha gained, 54,001 ha stable, on cells that are the same size everywhere, with a transcript a verifier replays without seeing either mask. The first thing this diff found was a processing artifact in our own pipeline.



Transcript: 142 kB, verifies True; with the operation swapped it verifies False. (The dates mix season and disturbance; the claim here is comparability, not the trend.)

8. Looking is not seeing: the observation ledger

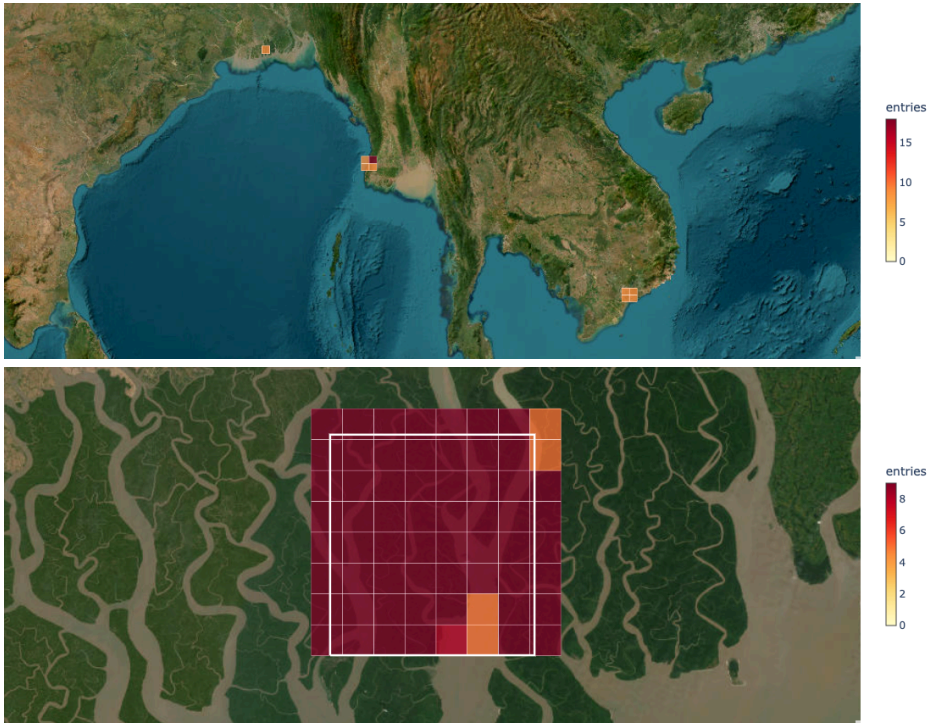
> Takeaway. Every scene ever selected for this site, most recent first, contributes the cells no later scene saw: a best-available-pixel composite whose provenance is 1 root rather than a paragraph. 100 % of the area has been seen at all; the rest is itself a root you can query. Cloud masking is one more mask in the same algebra.



Left: which observation each cell's best pixel comes from. Right: how many observations touched each cell, the revisit density.

9. A catalogue that indexes roots

> Takeaway. 36 entries across four sites, 21 distinct roots, and the index never sees a coordinate. Ask what covers Sundarbans? and every hit comes with two exact fractions a bbox search cannot give. Roll the entries up and count, and the catalogue draws where the effort is.



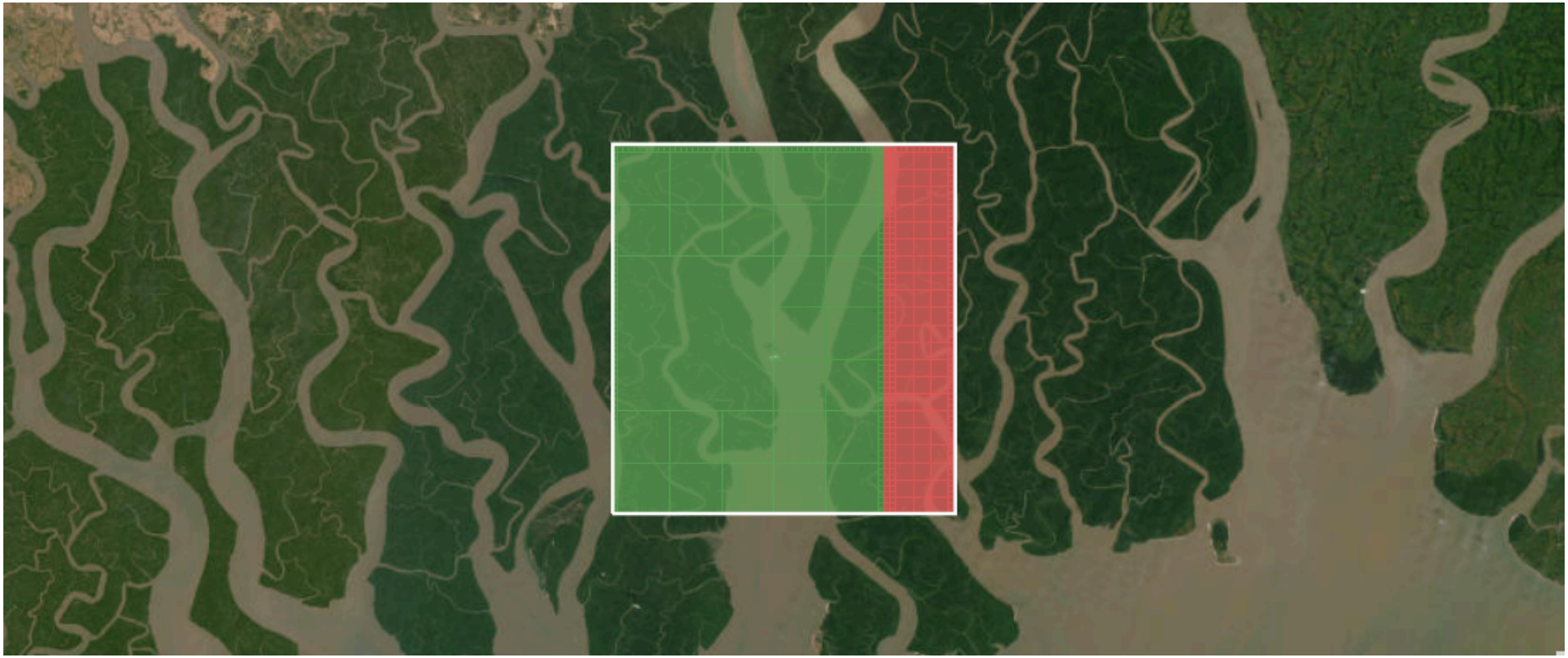
entry	lineage	date	relation	covers % of my ground	% of it in my ground
sundarbans/S2A_45QYE_20200516_0_L2A/valid	Sentinel-2 valid data	2020-05-16	equal	100.0	100.0
sundarbans/S2B_45QYE_20171223_0_L2A/valid	Sentinel-2 valid data	2017-12-23	equal	100.0	100.0
sundarbans/S2B_45QYE_20201107_1_L2A/valid	Sentinel-2 valid data	2020-11-07	equal	100.0	100.0
sundarbans/S2B_45QYE_20241216_0_L2A/valid	Sentinel-2 valid data	2024-12-16	equal	100.0	100.0
sundarbans/aoi	study area bbox		equal	100.0	100.0
sundarbans/S2B_45QYE_20171223_0_L2A/mangrove	mangrove mask (NDVI)	2017-12-23	within	66.1	100.0
sundarbans/S2B_45QYE_20201107_1_L2A/mangrove	mangrove mask (NDVI)	2020-11-07	within	65.6	100.0
sundarbans/S2B_45QYE_20241216_0_L2A/mangrove	mangrove mask (NDVI)	2024-12-16	within	65.3	100.0

2 distinct lineages have outputs over this ground (Sentinel-2 valid data, mangrove mask (NDVI)). Same ground, different lineage is the consilience query.

Left: entries per 1,440 km² cell across the region. Right: the same count at site scale. A group that does not want to publish an exact footprint can publish its coarse cells and still appear here.

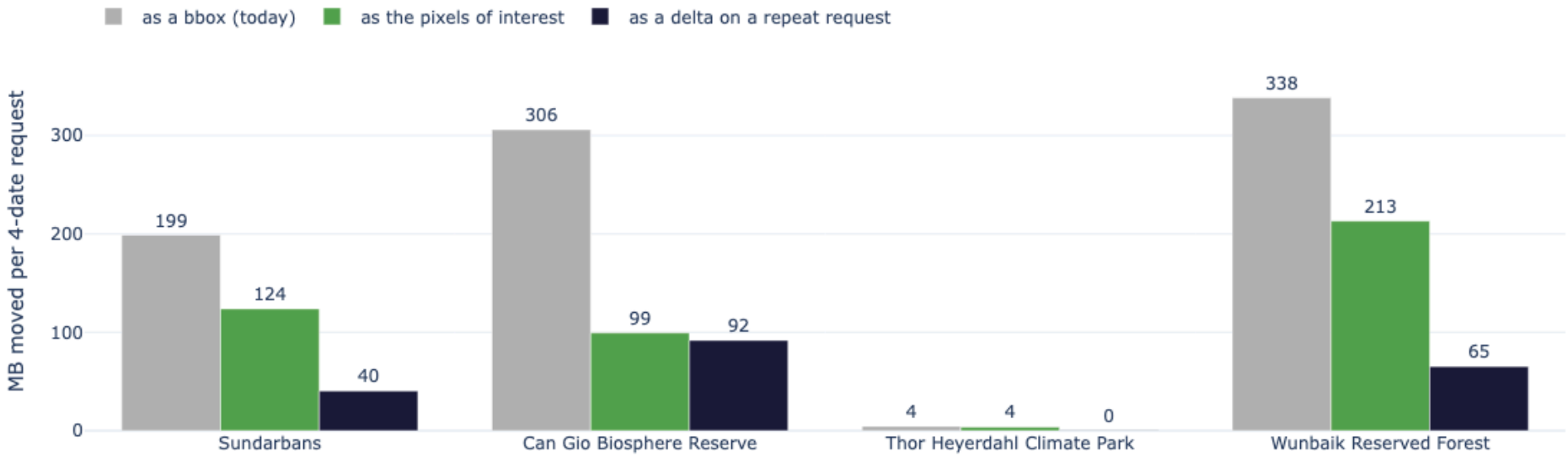
10. What the platform gains: execute only new ground

> Takeaway. A job is its workflow hash plus the root of the ground it computed. This request is one query against the ledger of 16 prior runs: 80 % already computed, so the decision is to execute the 20 % that is new, serve the rest. Compute stops scaling with how many people ask and scales with how much ground is new.



11. Cloud in, cloud out: request the ground, not the box

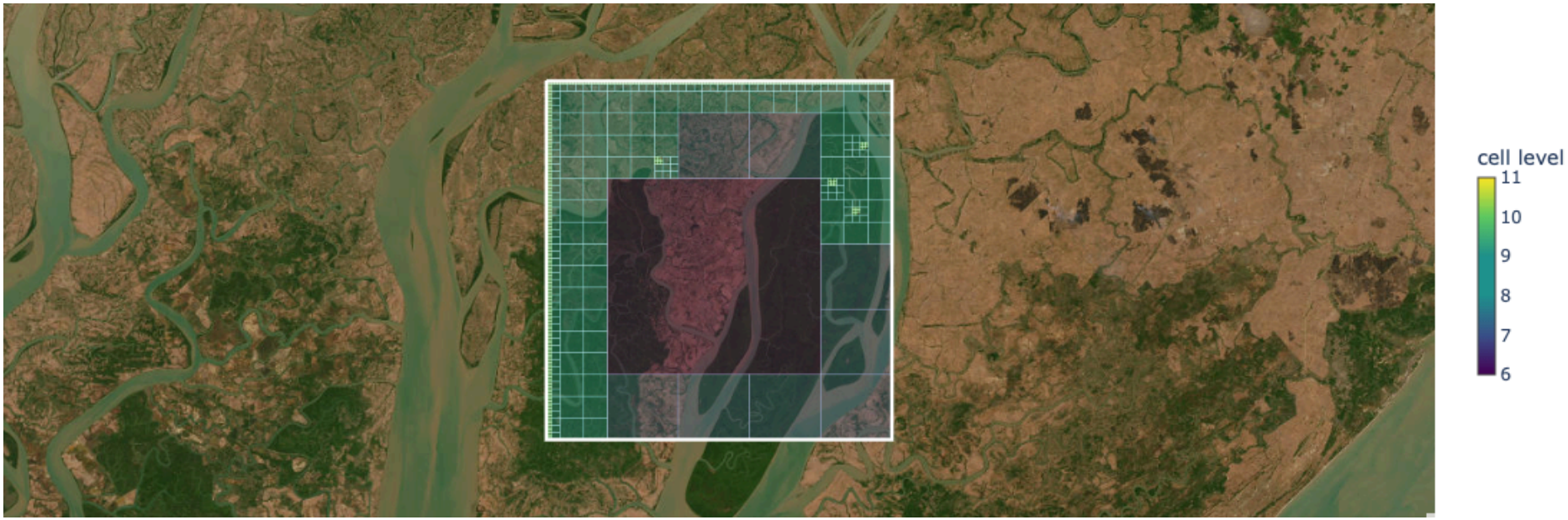
> Takeaway. Today the operator answers a request with a bounding box and every byte in it moves, needed or not. Stated as cells of interest, a request moves 48 % fewer bytes; as a delta on a repeat, 77 % fewer. At 500 TB of annual egress and $0.09/GB that is $21,624 and $34,482 a year. Both sides hold the same root, so the reply is also a receipt.



Bars are this repository's own sites: four dates, three bands at 10 m, all counted in the same equal-area zones. The sliders above set the price and scale; the fractions hold at any scale.

12. What travels with the number

> Takeaway. The application package plus one CWL step at the end, and the STAC item carries the Fingerprint extension in OGC API - DGGS words: the item's fingerprint is the root of its geometry's compact zone list on the OGC-registered rHEALPix DGGRS at refinement level 11; the mask asset carries the fingerprint of the ground it detected; both zone lists travel as assets. Verified here from the JSON alone (geometry, fingerprint, mangrove_mask_tif_fingerprint, mangrove_mask_tif (recomputed)): the same roots on this laptop, in the Linux container, and in the Rust test suite, to the byte.



13. The ask

> Three packages, each one file. A STAC extension of three fields, so ground becomes a primary key (stac-extension/). A coverage index behind the catalogue, so overlap becomes a fair fraction and jobs execute only new ground (burin_core.Index). One platform-level step after every execution (cwl/fingerprint.cwl), so all of this appears without any workflow author changing anything. STAC gives the primitive a ledger and a transport. The primitive gives STAC an identity for ground.